



Effets des inhibiteurs SGLT2 sur la pression artérielle et les paramètres anthropométriques dans le syndrome métabolique.

MOSTEFAOUI Leila⁽¹⁾, HADJARI Mohammed⁽²⁾, Khaladi Hadj Mostafa⁽²⁾.

(1)Research Laboratory on Biological Systems and Geomatics (LRSBG), Faculty of Natural and Life Sciences, Mustapha Stambouli University of Mascara, Algeria ; 29000, Algeria.

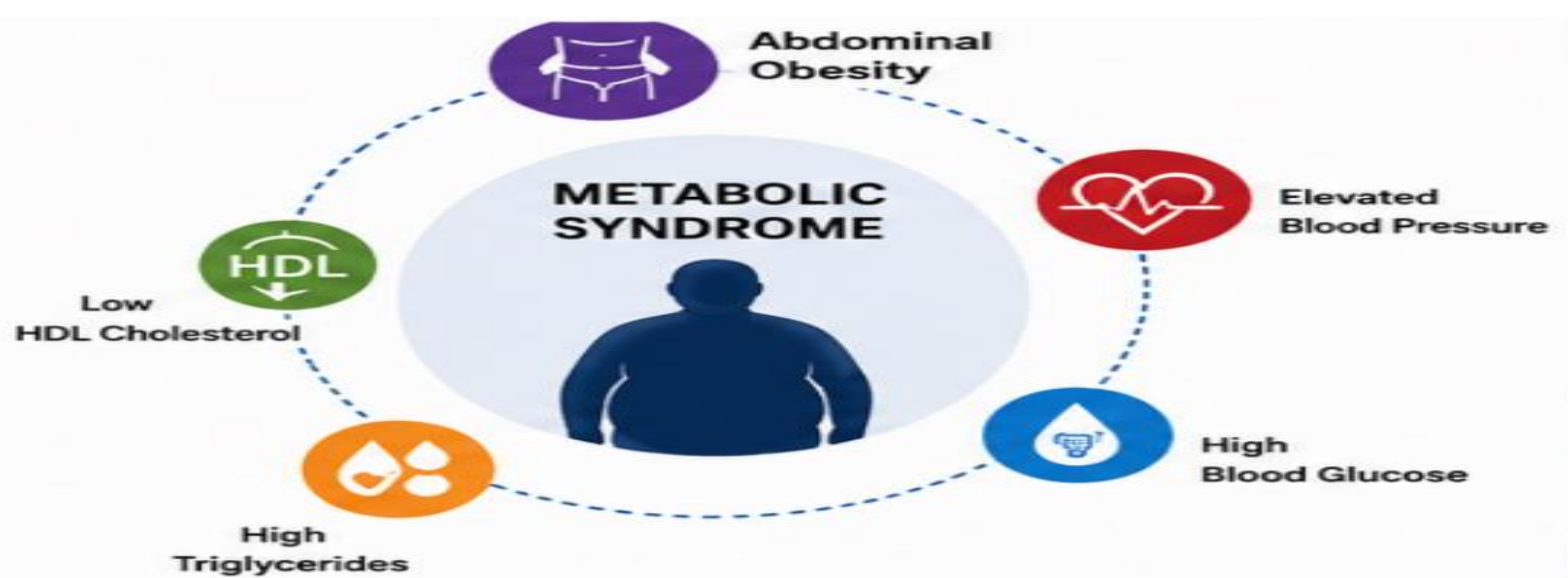
(2)Technological Laboratory for Artificial Intelligence and Food Security (LABTEC-AI), Research Unit on Food Safety and Contaminant Monitoring (FSCM), Department of Biology, Faculty of Sciences of Nature and Life, Mustapha Stambouli University, Mascara 29000, Algeria.

INTRODUCTION

Metabolic syndrome is a major public health concern due to its strong association with type 2 diabetes mellitus and cardiovascular disease. It is characterized by a cluster of metabolic abnormalities including abdominal obesity, hyperglycemia, hypertension, and dyslipidemia [1,2].

Sodium-glucose cotransporter-2 inhibitors (SGLT2 inhibitors) have emerged as an innovative therapeutic class that improves glycemic control while providing additional cardiovascular and metabolic benefits [3,4].

The aim of this study was to evaluate the impact of SGLT2 inhibitors on glycemic control and cardiovascular risk factors in Algerian patients with metabolic syndrome.



Matériel et méthodes

Étude comparative multicentrique

Population : 600 patients algériens présentant un syndrome métabolique

Répartition des groupes (n = 150 par groupe)



G1

Metformine
(MET)



G2

iSGLT2
(Dapagliflozin 10 mg/l)



G3

Association
iSGLT2 + Metformine



G4

Témoins sains
(non diabétiques)

Paramètres évalués :

Anthropométriques

- Indice de masse corporelle (IMC)
- Tour de taille (TT)
- Rapport taille/hanche (RTH)



Hémodynamiques

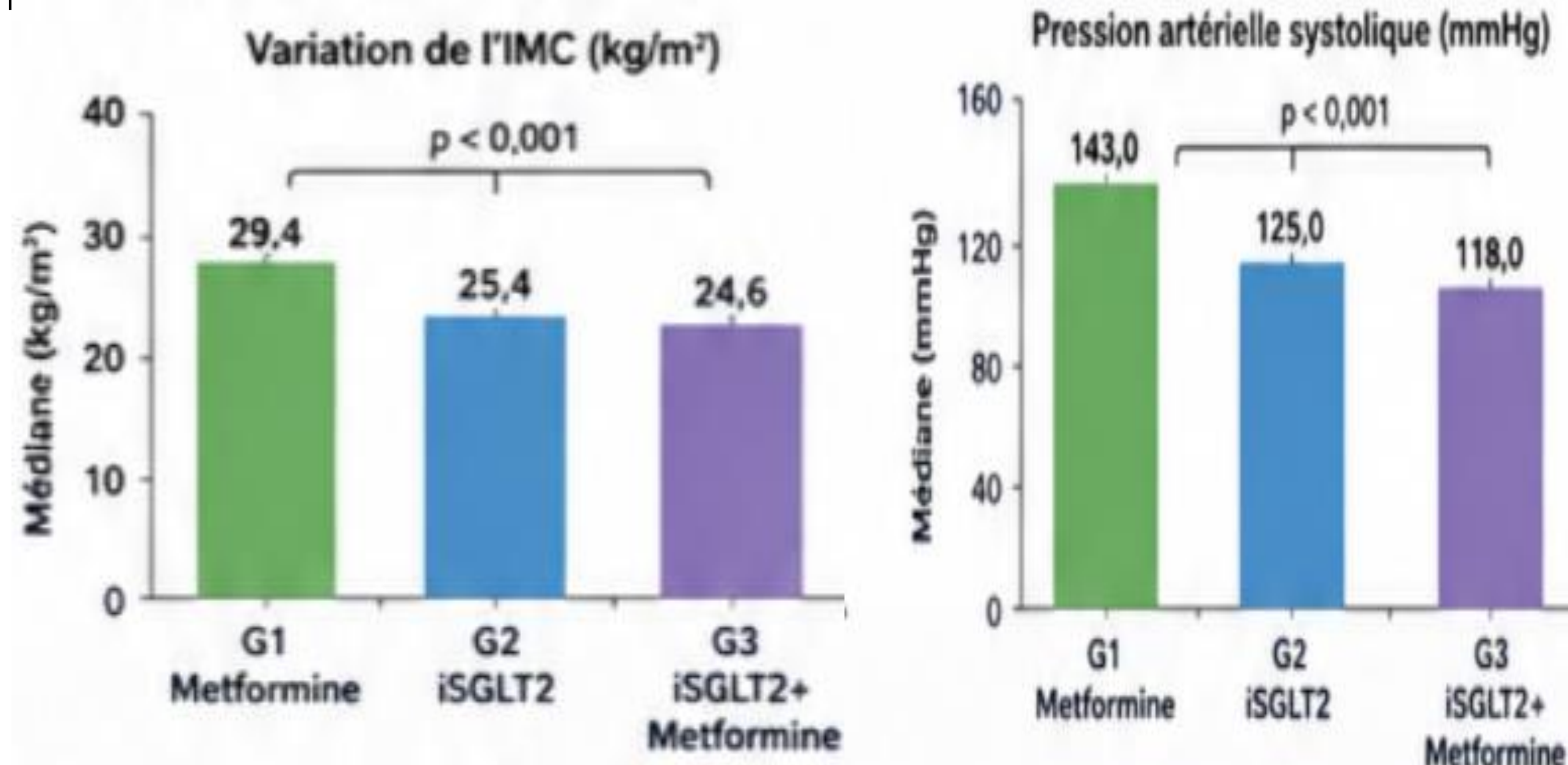
- Pression artérielle systolique (PAS)
- Pression artérielle diastolique (PAD)



Analyse statistique : ANOVA à un facteur avec test post-hoc de Tukey.
Différence significative si $p < 0,05$.

RESULTS & DISCUSSION

The findings demonstrated a favorable effect of SGLT2 inhibitor-based therapy on glycemic control and several cardiometabolic parameters. Patients receiving SGLT2 inhibitors showed improvements in their metabolic and cardiovascular risk profiles, suggesting benefits beyond glucose lowering. Positive changes were observed in anthropometric measures and blood pressure parameters, supporting a broader cardiometabolic benefit of SGLT2 inhibitor therapy. These findings are consistent with large international clinical trials reporting reductions in cardiovascular risk and improvements in metabolic outcomes among patients treated with SGLT2 inhibitors [3–6]. Overall, the combined improvement in glycemic control and cardiovascular risk factors supports the growing role of SGLT2 inhibitors in the management of patients with metabolic syndrome and type 2 diabetes.



Les patients traités par iSGLT2, seuls ou en association avec la metformine, présentent une réduction significative du poids, de l'IMC, du tour de taille, du RTH ainsi que de la pression artérielle systolique et diastolique comparés au groupe metformine seule ($p < 0,001$).

CONCLUSION

SGLT2 inhibitors appear to be a promising therapeutic strategy for improving glycemic control and cardiometabolic health in patients with metabolic syndrome. Their beneficial effects on anthropometric parameters and blood pressure may contribute to reducing overall cardiovascular risk in high-risk populations.

REFERENCES

- [1] International Diabetes Federation. IDF Diabetes Atlas, 11th Edition. Brussels: IDF; 2025.
- [2] American Diabetes Association. Standards of Care in Diabetes—2025. Diabetes Care. 2025.
- [3] Zinman B, Wanner C, Lachin JM, et al. Empagliflozin, Cardiovascular Outcomes, and Mortality in Type 2 Diabetes. N Engl J Med. 2015;373:2117–2128.
- [4] Neal B, Perkovic V, Mahaffey KW, et al. Canagliflozin and Cardiovascular Outcomes in Type 2 Diabetes. N Engl J Med. 2017;377:644–657.
- [5] McMurray JJV, Solomon SD, Inzucchi SE, et al. Dapagliflozin in Patients with Heart Failure and Reduced Ejection Fraction. N Engl J Med. 2019;381:1995–2008.
- [6] Verma S, McMurray JJV. The Serendipitous Story of SGLT2 Inhibitors in Heart Failure. Circulation. 2018;139:2537–2541.